

```
import pandas as pd
import numpy as np

df = pd.read_csv("Students.csv")
print(df, "\n")

print(df.describe(include='all'), "\n")

print(df['age'], "\n")

df.rename(columns={'age': 'student_age'}, inplace=True)
print(df.columns, "\n")

df['marks'] = pd.to_numeric(df['marks'], errors='coerce')
mean_marks = df['marks'].mean()
df['marks'] = df['marks'].fillna(mean_marks)

median_age = df['student_age'].median()
df['student_age'] = df['student_age'].fillna(median_age)

df = df.drop_duplicates()
print(df, "\n")

print(df.head(5), "\n")
print(df.tail(5), "\n")
print(df.sample(3), "\n")

df_cleaned = df.dropna()
print(df_cleaned, "\n")

df_cleaned.to_csv("Cleaned_Students.csv", index=False)
```

```
===== RESTART: C:/Users/krsb/OneDrive/Desktop/internship/DAY 16 TASK.py =====
```

	id	name	age	marks	city
0	1	Asha	21.0	85	Mysore
1	2	Ravi	22.0	90	Bangalore
2	3	John	NaN	78	Mangalore
3	4	Meena	21.0	abc	Delhi
4	5	Kiran	23.0	88	Chennai
5	6	Ravi	22.0	90	Bangalore
6	7	Pooja	NaN	95	Mysore
7	8	Arun	24.0	70	Hyderabad
8	3	John	NaN	78	Mangalore

	id	name	age	marks	city
count	9.000000	9	6.000000	9	9
unique	NaN	7	NaN	7	6
top	NaN	Ravi	NaN	90	Mysore
freq	NaN	2	NaN	2	2
mean	4.333333	NaN	22.166667	NaN	NaN
std	2.345208	NaN	1.169045	NaN	NaN
min	1.000000	NaN	21.000000	NaN	NaN
25%	3.000000	NaN	21.250000	NaN	NaN
50%	4.000000	NaN	22.000000	NaN	NaN
75%	6.000000	NaN	22.750000	NaN	NaN
max	8.000000	NaN	24.000000	NaN	NaN

```
0    21.0
1    22.0
2     NaN
3    21.0
4    23.0
5    22.0
6     NaN
7    24.0
8     NaN
```

```
Name: age, dtype: float64
```

```
Index(['id', 'name', 'student_age', 'marks', 'city'], dtype='object')
```

	id	name	student_age	marks	city
0	1	Asha	21.0	85.00	Mysore
1	2	Ravi	22.0	90.00	Bangalore
2	3	John	22.0	78.00	Mangalore
3	4	Meena	21.0	84.25	Delhi
4	5	Kiran	23.0	88.00	Chennai
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>>> |